Solution :-

Sol^n is the homogenous mixture of 2 or more component.

Solution = Solute + Solvent

↓ Solute: less in amt. (moles)

↓ Solvent: excess in amt. (in terms of moles)

180 g Urea + 90 g water

→ 3 mol → 5 mol

↳ solvent

→ 300 g urea + 684 g Sucrose + ~~180~~ 180 g water

→ 5 mol → 2 mol 10 mol

Sol^n component	Solute	Solvent
2	1	1
3	2	1
4	3	1

always = 1

- Nature of sol^n is same as that of solvent.

Sol^n
- → solid
- → liquid
- → Gas

depend upon nature of solvent

sol^n
- liq + liq → water + alcohol
- gas + gas → air
- solid + solid → alloys
- liq. + gas → cold drink
- liq. + solid → Nacl + water

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Concentration Terms :-

1. Molarity (M)
2. Normality (N)
3. Molility (m)
4. Formality (F)
5. Strength (S)
6. Mole fraction
7. % → w/w %
   → ~~V~~ v/v %
   → w/v %
8. PPM

1. Molarity (M) :- no. of moles of solute ~~+~~nt in 1L soln.

$$M = \frac{n_{solute}}{V_{soln}(L)} \qquad \boxed{M = \frac{n}{V}} \qquad \text{Unit} = \text{mol } L^{-1}$$

M

2. Normality (N) : It is no. of g. eq. of solute ~~+~~nt in 1L soln

$$N = \frac{\text{no. of g. eq. of solute}}{V_{soln}(L)} \qquad \text{Unit: eq } L^{-1}$$

N

$$\text{no. of g. eq.} = \frac{wt}{Eq\ wt} = \frac{w}{E} = \frac{w}{M/n\text{-factor}} = \frac{w}{M} \times n\text{-factor} = n \times n\text{-factor}$$

$$\text{no. of g. eq.} = N \times V = \frac{w}{E} = n \times n\text{-factor}$$

$$N = \frac{n \times n\text{-factor}}{V_{soln}(L)}$$

$$N = M \times n\text{-factor}$$

Note: First calculate Molarity then calculate Normality

3. Molality : It is moles of solute ~~+~~nt in 1 Kg solvent.

$$m = \frac{\text{moles of solute}}{\text{wt. of solvent (in 1Kg)}}$$

$$m = \frac{n_{solute}}{W_{solvent}(kg)}$$

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It does not depend upon Temp.

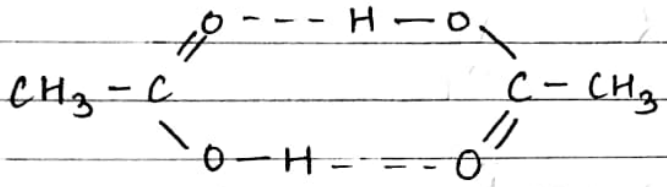
4. Formality (F) : It is no. of formula mass in 1L soln
→ defined for ionic compd.

$$F = \frac{\text{M solute}}{(\text{Formula mass}) \times V_{soln}(L)}$$

20 g NaOH in 2L → Fw = 40 $\quad M = \frac{20/40}{2} = 1/4$

Mw = 40 $\quad F = \frac{20}{40 \times 2} = 1/4$

when Fw = Mw $\quad F = W$



Fw = 120

6g CH_3COOH in 1L soln $\quad M = ? \quad F = ?$

$$M = \frac{6/60}{1} = 0.1\,M$$

$$N = \frac{6}{120 \times 1} = 0.05$$

Que 44.8 g benzoic acid is +nt in 2L soln Find Molarity & formality ?

$C_6H_5COOH = 7 \times 12 + 2 \times 16 + 6$

$= 84 + 32 + 6 = 84 + 38 = 122$

$$M = \frac{\frac{44.8}{122} \times 10}{2} = \frac{2}{10 \times 2} = 0.1\,M$$

$$F = \frac{44.8}{122 \times 2} = \frac{2}{1 \times 2 \times 10} = 0.1$$

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Strength : wt. of solute in 1 L soln

$S = \frac{\text{wt. of solute}}{V_{soln}\ (L)}$ Unit : g/L

Mole fraction : (X)

A	B	C
n_A	n_B	n_C

$X_A = \frac{n_A}{n_T}$ $X_A + X_B + X_C = 1$

$X_B = \frac{n_B}{n_T}$

$n_T = n_A + n_B + n_C$

$X_C = \frac{n_C}{n_T}$

Soln $X_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$

$X_{solvent} = \frac{n_{solvent}}{n_{solute} + n_{solvent}}$

$X_{solute} + X_{solvent} = 1$

$\frac{X_A}{X_B} = \frac{n_A}{n_B}$

7. $\%\ w/w = \frac{\text{wt. of solute}}{\text{wt. of soln}} \times 100$

20 % w/w Urea = 20 g urea in 100 g soln

$\%\ V/V = \frac{\text{Vol. of solute}}{\text{Vol. of soln}} \times 100$

20 % V/V NaOH = 20 ml NaOH in 100 ml soln

$\%\ W/V = \frac{\text{wt. of solute (g)}}{\text{Vol. of soln (ml)}} \times 100$

40% W/V H_2SO_4 soln → 40 gm H_2SO_4 in 100 ml soln

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Ques 20% by wt NaOH soln Find molality ?

20 gm NaOH +nt in 100 gm soln

$$\text{molality} = \frac{20/40}{100/1000} = \frac{1/2}{1/10} = \frac{10}{2} = 5m$$

Ques 20% W/V NaOH soln find Molarity ?

20 gm NaOH +nt in 100 ml soln

$$\text{Molarity} = \frac{20/40}{100/1000} = \frac{1/2}{1/10} = 5M$$

PPM : Parts Per million

Very dilute soln

$$\text{PPM (by wt)} = \frac{\text{wt. of solute}}{\text{wt. of soln}} \times 10^6$$

$$\text{PPM (by Vol.)} = \frac{\text{Vol. of solute}}{\text{Vol. of soln}} \times 10^6$$

ppb : Parts Per Billion

$$\text{ppb (by wt)} = \frac{\text{wt. of solute}}{\text{wt. of soln}} \times 10^9$$

AIPMT

Ques How many gram of dibasic acid (Mol. wt. = 200) should be +nt 100 ml of aq. soln to give 0.1 N soln ?

$$N = \frac{W/M_w \times n\text{-factor}}{V_{sol^n}}$$

$$\frac{N \times V \times M_w}{n\text{-factor}} = W$$

$$\frac{0.1 \times 100 \times 200}{1000 \times 2} \cancel{10} = W = 1\ gm$$

AIPMT-95,97

Ques Which of the following conc^n term is independent of Temp. ?

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① Molarity ② Molality ③ Normality ~~④ Formality~~

Ques Calculate mass of sugar & water req.d to prepare 250 gm of 20% w/w soln.

~~20 gm sugar +nt in 100 gm soln~~

$$\frac{20}{5} = \frac{Wt\ solute}{250} \times \frac{100}{5}$$

$$\frac{250}{5} = W\ solute = 50\ gm$$

$$W\ solvent = 250 - 50 = 200\ gm$$

AIPMT-07

Ques Conc. of sulphuric acid is 98% H_2SO_4 by mass & has a density of $1.8\ g\ L^{-1}$. Vol of acid req.d to ~~mass~~ make one litre of 0.1 M H_2SO_4 soln

① 11.10 ml ② 16.65 ml ③ 22.20 ml ④ 5.55 ml

98 gm H_2SO_4 +nt in 100 gm soln

$$D = \frac{M}{V}$$

$$1.8 = \frac{98}{V}$$

$$V = \frac{980}{18} = 5$$

$$0.1\ M = \frac{n_{H_2SO_4}}{1} \Rightarrow n_{H_2SO_4} = 0.1\ mol$$

$$V_{H_2SO_4} = 0.1 \times 22.4$$

$$= 2.24$$

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Ques Calculate Molarity of 400 g pure water if $d_{H_2O} = 1 g/ml$

$V = \frac{400}{1} = 400 ml$

$M = \frac{400/18}{400} \times 1000 = \frac{400}{400 \times 18} \times 1000 = 55.5$

* Ques Calculate Molarity of a sol^n of $MgCl_2$ if 400 ml $MgCl_2$ have 3.011×10^{22} chloride ions

$MgCl_2 \rightarrow Mg^{2+} + 2Cl^-$

0.025 0.025 0.05 mol

$M = \frac{25 \times 10^{-3}}{400} \times 1000 =$

AIPMT-05 IAS

Ques Calc. mole fraction of solute in one molal aq. sol^n?

1 molal → 1 mol of solute + nt in 1 kg solvent

$n_{H_2O} = \frac{1000}{18} = 55.5$

$X_{solute} = \frac{1}{1 + 55.5} = \frac{1}{56.5}$

$= \frac{10}{565} = 0.0177$

NCERT

Ques Calc. mass of urea reqd in making 5 Kg 0.25 m aq. sol^n

$m = \frac{\text{moles of solute}}{\text{wt (Kg)}} = \frac{w/60}{wt}$

$0.25 \times 5 \times 60 = w$

$w = 15.00 \times 5 = 75 \times 1000$

$= 75000 gm.$

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$39 + \cancel{108}\ 127 = 166$

Ques Calculate ① Molarity ② Molality ③ Mole fraction of KI if density 20 % w/w aq. KI is 1.202 g/ml

20 gm KI + nt in 100 gm soln

$\rho = 1.202$

① $M = \frac{n}{V} = \frac{20/166}{83.3} \times 1000$

$= \frac{20}{166 \times 83.3} \times 1000 = 1.44$

$V = \frac{M}{\rho} = \frac{100}{1.202}$

$= \frac{10000}{120.2}$ ml

② $m = \frac{20/166}{80} \times 1000 = \frac{\cancel{20}}{4\ \cancel{80} \times 166} \times 1000 = 1.50$

③ $X_{\cancel{*}solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$

$n_{solute} = \frac{20}{166} = 0.120$

$= \frac{0.12}{0.12 + 4.44} = 0.026$

$n_{solvent} = \frac{80}{18} = 4.44$

Ques Eq. moles of urea & water are mixed to prepare an aq. soln what is w/w % of urea & also find Molality.

$W/W\ \% = \frac{\cancel{n} \times \cancel{60}\ 10^5}{\cancel{2n} \times \cancel{78}\ 13} \times 100 = \frac{500}{13}\ \%$

$m = \frac{n \times 1000}{n \times 18} = \frac{1000}{18}$

$= 55.5\ m$

Ques Calc. Molarity & molality of 25 % w/w H_2SO_4 soln if sp. gravity of soln is 1.2 g/ml

25 gm H_2SO_4 + nt in 100 gm soln

$M = \frac{25/98 \times 1000}{\frac{100}{1.2}} = \frac{25}{98} \times 1.2 \times 1000$

$= \frac{25 \times 12 \times 100}{100 \times 98}$

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$$M = \frac{300}{98} \approx 3\,M$$

$$\text{Molality} = \frac{25/98 \times 1000}{75} = \frac{25 \times 1000}{98 \times 75 \times 3}$$

$$= \frac{1000}{294} = 3.4\,m$$

AIPMT

Ques 25.3 g of Na_2CO_3 is dissolved in enough water to make 250 ml of soln. If Na_2CO_3 dissociates completely then molar conc^n of Na^+ & CO_3^{2-} ions are respectively (Given Mw. of Na_2CO_3 = 106 g/mol)

(a) 0.955 M & 1.910 M
(b) 1.910 M & 0.955 M
(c) 1.90 M & 1.910 M
(d) 0.477 M & 0.477 M

$Na_2CO_3 \rightarrow 2Na^+ + CO_3^{2-}$

C 0 2C C

$$C = \frac{25.3/106 \times 1000}{250}$$

$$= \frac{25.3 \times 1000}{106 \times 250} = \frac{50.6}{53}$$

Ques Calculate amt. of oxalic acid req^d to obtain 250 ml of semimolar soln?

$H_2C_2O_4 \rightarrow 2 + 64 + 24 = 88 + 2 = 90$

$$1/2 = \frac{W/90 \times 1000}{250}$$

$$\frac{1}{2} = \frac{W}{90} \times 4$$

$$\frac{45}{4} = W = 11.25\ gm$$

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Volume Strength of H_2O_2 :- The volume of O_2 produced by decomposition of unit volume of H_2O_2 at NTP/STP

Relation b/w vol. strength of H_2O_2, M, N & W/V % :-

Let vol. strength of H_2O_2 is 'X' V

→ 1 L H_2O_2 will produce 'X' L O_2 at STP

$$\underset{2\text{ mol}}{2H_2O_2} \longrightarrow \underset{1\text{mol}}{2H_2O + O_2} \longrightarrow \frac{X}{22.4} \text{ mol of } O_2$$

1 mol is of O_2 is obtained by decomposition of = 2 mol H_2O_2

$\frac{X}{22.4}$ mol ⟶ $= \frac{X}{22.4} \times 2$

$= \frac{X}{11.2}$ mol of H_2O_2

① Molarity $= \frac{\text{no. of moles}}{\text{Vol (L)}}$

$$M = \frac{\frac{X}{11.2}}{1} = \frac{X}{11.2} \text{ mol } L^{-1}$$

$$M = \frac{X}{11.2} \quad *$$

② density $M = \frac{n}{V} = \frac{w}{Mw \times V} = \frac{d}{Mw}$

$$d = M \times Mw$$

$$d = \frac{X}{11.2} \times 34$$

③ Normality $N = M \times n\text{-factor}$

$$= \frac{X}{11.2} \times 2$$

$$N = \frac{X}{5.6} \quad *$$

④ Strength : $S = N \times E$

$$S = \frac{X}{5.6} \times 17 \text{ g/L} \qquad E = \frac{34}{2} = 17$$

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(5) % W/V $= \frac{X}{5.6 \times 1000} \times 17 \times 100$

$= \frac{X}{5.6} \times \frac{17}{10}$

Ques Find Vol. strength of 1.5 N H_2O_2 sol^n ?

$N = \frac{X}{5.6}$

$1.5 \times 5.6 = X = 8.40 = 8.4\ L$

Ques Calculate strength of 44.8 V H_2O_2 sol^n in terms of ① Molarity ② Normality ③ Strength ④ % W/V

① Molarity $= \frac{X}{11.2} = \frac{44.8}{11.2} = 4$

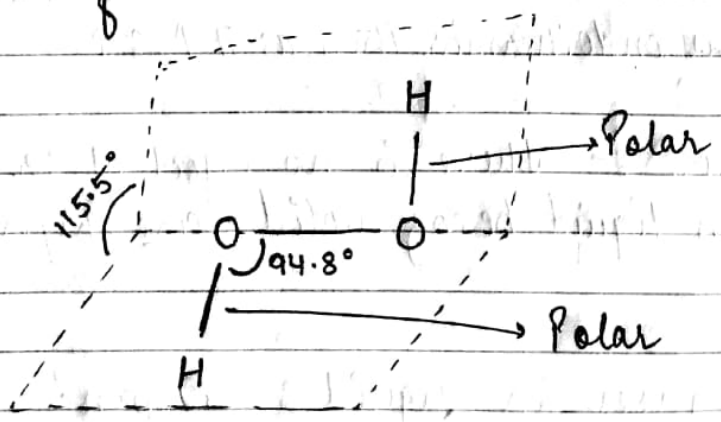
② Normality $= \frac{X}{5.6} = \frac{44.8}{11.2} = 8$

③ Strength = N X E

$= \frac{X}{5.6} \times 17\ g/L = 8 \times 17 = 136\ g/L$

④ % W/V $= \frac{X}{5.6} \times \frac{17}{10} = \frac{136}{10} = 13.6$

Structure of H_2O_2 :-

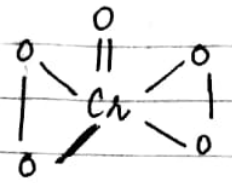


Test of H_2O_2

↳ H_2O_2 reacts with Titanic acid $Ti(OH)_4$

$H_2O_2 + H_2SO_4 + K_2Cr_2O_7 \longrightarrow CrO_5$

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Cr^{6+}

Butterfly str.

Solubility :- The max^m amt. of solute dissolved in a specified amt of solvent at const. Temp.

Solubility depends upon

1. Nature of solute
2. Nature of Solvent
3. Temp.
4. Pressure

Solubility Principle :: Like dissolve like

Solubility of solids in liquid : Polar solute are soluble in polar solvent & non-polar solute are soluble in non-polar solvent.

$$\text{Solute} + \text{Solvent} \underset{\text{crystallisation}}{\overset{\text{dissolution}}{\rightleftharpoons}} \text{Solution}$$

Effect of Temp. :-

If dissolution process is exothermic then as T↑ Solubility↓

If dissolution process is endothermic then as T↑ S↑

Effect of Pressure :- There is no effect of Pr. on solubility of solid in liquid becoz solid are highly incompressible.

Solubility of gases in liquid :- Depends upon

Nature of Gas :

eg NH_3 & HCl are highly soluble then O_2, N_2 & He are slightly soluble.

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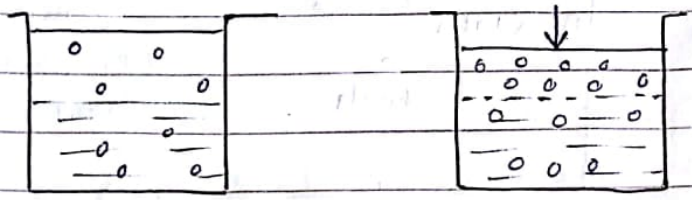
Temperature :-

Dissolution of gas in liquid is an endothermic process like condensation (Gas → Liq.) as Temp ↑ Sol. ↓

eg : aquatic animals remains in cold water

Reason : Gas dissolution ↑

Pressure : Pr. ↑ Sol. ↑



Henry's Law : It states that at constant temp. solubility of a gas in liquid is directly proportional to pressure of gas.

OR

Mole fraction of a gas in sol^n is directly proportional to Partial Pressure of the gas over sol^n.

OR

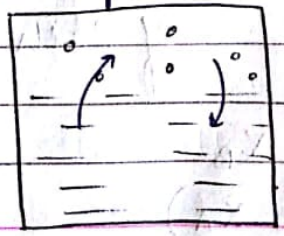
The partial Pressure of the gas in vapour Phase (P) is proportional to mole fraction of the gas in sol^n.

$P \propto X$

$P = K_H X$ $\quad K_H \rightarrow$ Henry's const

~~$X \propto \frac{1}{K_H}$~~ $\quad T \uparrow$ Sol. $\downarrow$ $X \downarrow$ $K_H \uparrow$

The Pressure exerted by vapour of liquid on its surface when liquid & vapours are in eq^m.



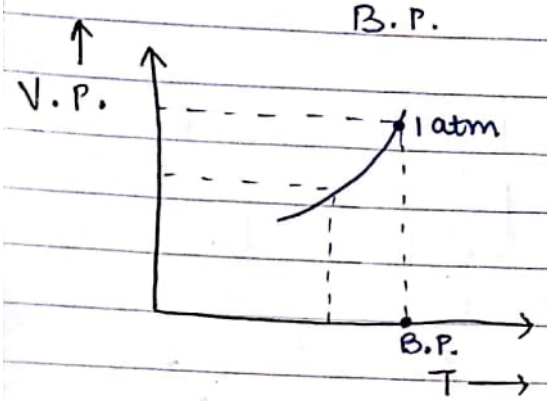
$H_2O(l) \rightleftharpoons H_2O(g)$

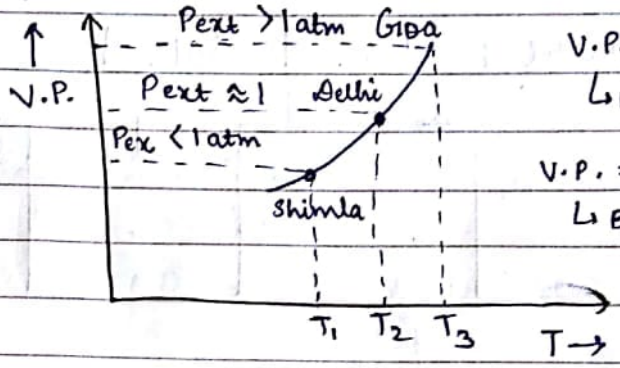
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Factors affecting Vapour Pressure :-

1. Nature of liq. :- More ~~vot~~ volatile liq. VP ↑
2. Temp. :

Boiling Point: The temp. at which V.P. becomes equal to atmospheric press. or ext. pr. is c/a





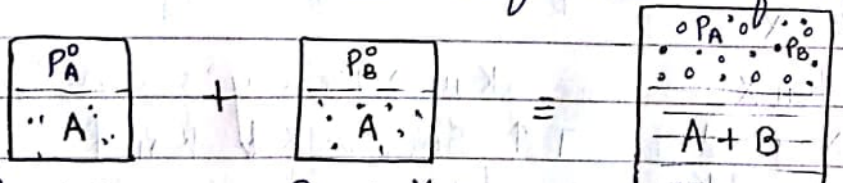
V.P. = 1 atm
↳ Normal B.P.
V.P. = Pext
↳ B.P.

Pex ↑	Pext ↓
V.P. ↑	V.P. ↓
B.P. ↑	B.P. ↓
Cooking time ↓	Cooking time ↑

V.P. of a Liquid - Liquid Solutions :-

(i.e. both liq. are volatile)

Raoult's Law :- The V.P. of a liquid is directly proportional to mole fraction of liq. in soln.



$P_A \propto X_A$ $\quad$ $P_B \propto X_B$

$P_A = P_A^\circ X_A$ $\quad$ $P_B = P_B^\circ X_B$

P_A → Partial V.P. of A
P_B → Partial " " B
P_A° → V.P. of Pure liq. A
P_B° → " " " " B

Total Pr. = Sum of Partial Pr. of each compd.

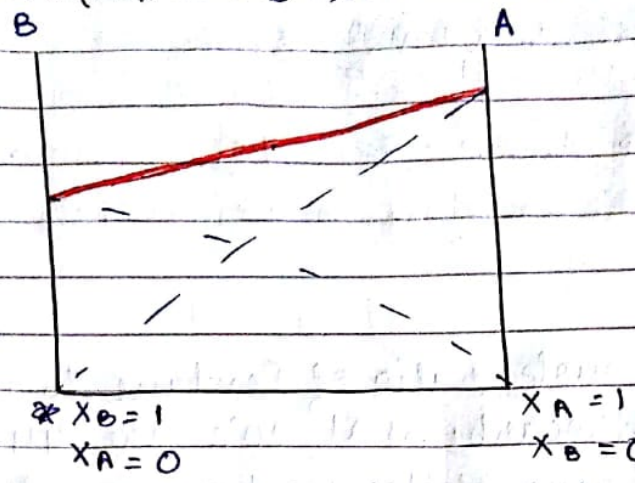
$$P_T = P_A + P_B$$

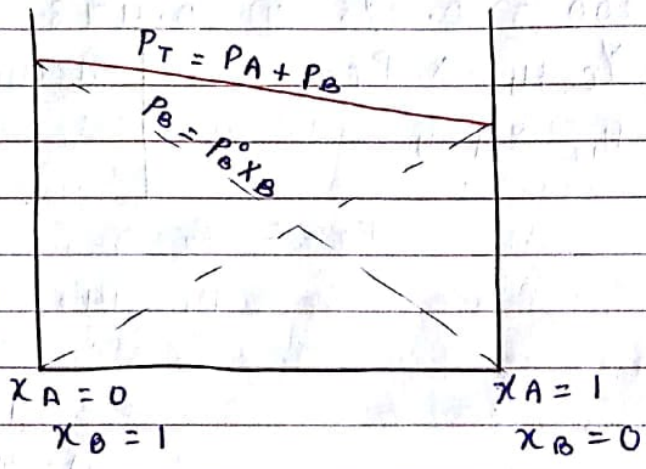
$$P_T = P_A^\circ X_A + P_B^\circ X_B$$

$$P_T = P_A^\circ X_A + P_B^\circ (1 - X_A)$$

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$P_T = (P_A^\circ - P_B^\circ) X_A + P_B^\circ$ $\quad A > B$





Mole fraction of A & B in vapour phase :-
Dalton's Law

Partial Pr. = mole fraction × P_T

$P_A = Y_A \times P_T$ $\quad$ $P_B = Y_B \times P_T$

$P_A^\circ X_A = Y_A \times P_T$ $\quad$ $P_B^\circ X_B = Y_B P_T$

$Y_A \rightarrow$ mole fraction of A in vapour phase
$Y_B \rightarrow$ mole fraction of B in vapour phase

Ques The V.P. of a liquid P & Q are 80 & 60 torr resp. The total V.P. of soln obtain by 3 moles of P & 2 moles Q would be.

① 140 torr ② 20 torr ③ 68 torr ④ 72 torr

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$$P_T = P_A^\circ X_A + P_B^\circ X_B$$

$$= \overset{16}{\cancel{80}} \times \frac{3}{\cancel{5}} + \frac{\cancel{60} \times 2}{11 \quad \cancel{5}}$$

$$= 48 + 22 = 70$$

Ques A sol^n has 1:4 mole ratio of Pentane to hexane. The V.P. of pure hydrocarbons at 20°C are 440 mm of Hg for pentane & 120 mm of Hg for hexane. The mole fraction of Pentane in vapour phase would be -

① 0.549 ② 0.200 ③ 0.786 ④ 0.478

$$P^\circ_{C_5H_{12}} X_{C_5H_{12}} = Y_{C_5H_{12}} \times P_T$$

$$\frac{\cancel{440}}{88} \times \frac{1}{\cancel{5}} = Y_{C_5H_{12}} \times 184$$

$$Y_{C_5H_{12}} = \frac{88}{184} \quad 2.$$

$$Y_{C_5H_{12}} = 0.478$$

$$\frac{n_{C_5H_{12}}}{n_{C_6H_{14}}} = \frac{1}{4}$$

$$P_T = \frac{\cancel{440}}{88} \times \frac{1}{\cancel{5}} + \frac{\cancel{120} \times 4}{\cancel{5}} \quad (24)$$

$$= 88 + 96$$

$$= 184$$

For solid - liq. sol^n :-

A → solvent (liquid) → volatile

B → solute (solid) → non-volatile

⇒ V.P. of liquid in pure state → P_A°

V.P. of solid in pure state → $\boxed{P_B^\circ = 0}$

$$P_T = P_A + P_B$$

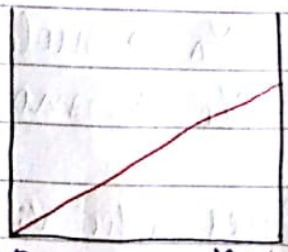
$$P_T = P_A^\circ X_A + \cancel{P_B^\circ X_B}^{\,0}$$

$$P_T = P_A^\circ X_A$$

$$\boxed{P_T = P_A^\circ \frac{N}{n+N}}$$

N → moles of solvent

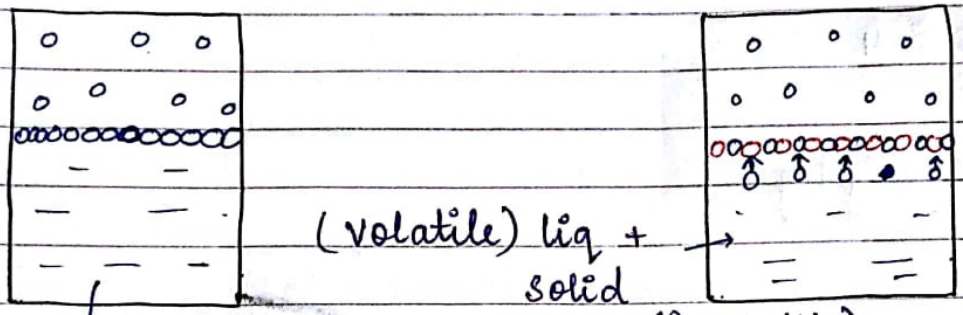
n → moles of solute



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→ In solid - liquid soln, V.P. of soln is directly proportional to mole fraction of solvent (i.e. amt. of volatile component)

When a non-volatile solute is added to volatile solvent then its vapour Pressure decrease because some surface is covered by non-volatile solute so less vapours are formed hence V.P. ↓ses.



n → mole of solute (non volatile)
N → moles of solvent (volatile)
P° → V.P. of pure solvent

$$P_T = P_A^\circ X_A + P_B^\circ X_B \to 0$$

$$P_T = P^\circ \otimes \frac{N}{n+N}$$

$$\frac{P_T}{P^\circ} = \frac{N}{n+N}$$

$$\frac{P^\circ}{P_T} = \frac{n+N}{N}$$

$$\frac{P^\circ}{P_T} = \frac{n}{N} + 1$$

$$\boxed{\frac{P^\circ - P_T}{P_T} = \frac{n}{N}}$$ → to find out wt. & molecular wt. of solute / solvent

$\boxed{P^\circ - P_T = \Delta P}$ → relative lowering V.P.

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Ques The vapour pressure at a given temp. of an ideal soln containing 0.2 mole of a non-volatile solute & 0.8 mole of solvent is 60 mm of Hg. Find vapour pressure of pure solvent at same T.

(a) 150 mm of Hg (b) 60 mm of Hg (3) 75 mm of Hg (4) 120 mm of Hg

$$\frac{P^\circ - P_T}{P_T} = \frac{n}{N}$$

$$\frac{P^\circ - 60}{60} = \frac{0.2}{0.8}$$

$$P^\circ - 60 = 15$$

$$P^\circ = 75 \text{ mm of Hg}$$

Ques Vapour Pressure of benzene at 30°C is 121.8 mm of Hg When 15 g of non-volatile solute is dissolved in 250 g of benzene its Vapour Pr. is used to 120.2 mm of Hg. Find Mw of solute.

(1) 356.2 (2) 456.3 (3) 350 530.1 (4) 656.7

$$\frac{121.8 - 120.2}{120.2} = \frac{15/Mw}{250/78}$$

C_6H_6

$72 + 6 = 78$

$$\frac{1.6}{120.2} = \frac{15 \times 78}{Mw \times 250}$$

$$Mw = \frac{234 \times 120.2}{50 \times 1.6}$$

$$= \frac{234 \times 120.2}{50 \times 1.6}$$

$$= 58.5 \times 6 = 351.0$$

Ques If V.P. of a soln a Glucose in water is 750 mm at 373 K. (760 mm of Hg.)

Find out (1) mole fraction of Glucose (2) Molality of soln

$\frac{1}{76}$

$P_T = P_A^\circ X_A + P_B^\circ X_B \rightarrow 0$

$750 = 760\, X_A$

$X_A = \frac{750}{760} = \frac{75}{76}$

$X_B = 1 - X_A = 1 - \frac{75}{76} = \frac{1}{76}$

(2) $m = \frac{\text{moles}}{\text{wt (kg)}} = \frac{\frac{1}{76}}{\frac{75}{76} \times \frac{18}{1000}} = \frac{1}{75} \times \frac{1000}{18}$

$= \frac{1}{75} \times 55.5$

Que V.P. of pure water is 112.5 mm of Hg at 300K. Find out the V.P. of 1 molal sol^n of a non-volatile solute in it ?

1 mol of solute + nt 1 Kg solvent $\hookrightarrow$ 1000g $\quad n = \frac{1000}{18} = 55.5$

$P_T = P_A^\circ X_A + P_B^\circ X_B \rightarrow 0$

$= 112.5 \times \frac{55.5}{1 + 55.5}$

$P_T = 112.5 \times \frac{55.5}{56.5}$

Que Calc. the wt. of non-volatile solute, should be dissolved in 78 g benzene to reduce its V.P. to 80% ?
(Given $M_{w_{\text{of solute}}} = 60$ g/mol)

$P^\circ = 100$

$P_T = 80$

$\frac{100 - 80}{80} = \frac{w/60}{78/78}$

$\frac{20}{80} = \frac{w}{60}$

$w = \frac{60}{4} = 15$ gm

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Ques The V.P. of liq. A & B are in ratio of $P_A^\circ : P_B^\circ = 1:2$ at certain temp., the mole ratio of A & B in liq. sol^n is 1:2, what is mole fraction of A in vapour phase?

$$\frac{n_A}{n_B} = \frac{1}{2} = \frac{X_A}{X_B}$$

$$\frac{P_A^\circ X_A}{P_B^\circ X_B} = \frac{Y_A \times P_T}{Y_B \times P_T}$$

$$\frac{1}{2} \times \frac{1}{2} = \frac{Y_A}{Y_B} = \frac{1}{4} = \frac{n'_A}{n'_B}$$

$$Y_A = \frac{1}{5}$$

Ques 2 liq. A & B form an ideal sol^n at 30°C. The V.P. of sol^n 2 mol of A & 3 mol of B is 600 mm of Hg. But if 1 moles of A & 4 moles of B are mixed then the V.P. of sol^n formed is 620 mm of Hg. What is V.P. of pure A & pure B?

$$P_T = P_A^\circ X_A + P_B^\circ X_B$$

$$600 = P_A^\circ \cdot 2/5 + P_B^\circ (3/5) - (1)$$

$$620 = P_A^\circ \, 1/5 + P_B^\circ \, 4/5 - (2)$$

$$\frac{30}{31} = \frac{1/5 \, (2P_A^\circ + 3P_B^\circ)}{1/5 \, (P_A^\circ + 4P_B^\circ)}$$

$$\frac{30}{31} = \frac{2P_A^\circ + 3P_B^\circ}{P_A^\circ + 4P_B^\circ}$$

$$30P_A^\circ + 120P_B^\circ = 62P_A^\circ + 93P_B^\circ$$

$$27P_B^\circ = 32P_A^\circ$$

$$\frac{27}{32} = \frac{P_A^\circ}{P_B^\circ}$$

Ques The V.P. of mix of 2 volatile compound A & B is $P = 254 - 119x$ mm of Hg (x = mole fraction of B in liq. sol^n) Find P_A° & P_B°

$$P_T = 254 - 119x$$

$$= P_A^\circ X_A + P_B^\circ X_B$$

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$P_T = P_A^\circ (1 - X_B) + P_B^\circ X_B$

$P_T = P_A^\circ + (P_B^\circ - P_A^\circ) X_B$

$P_T = P_A^\circ - (P_A^\circ - P_B^\circ) X_B$

$P_A^\circ = 254$ mm of Hg

$P_A^\circ - P_B^\circ = 119$

$254 - 119 = P_B^\circ$

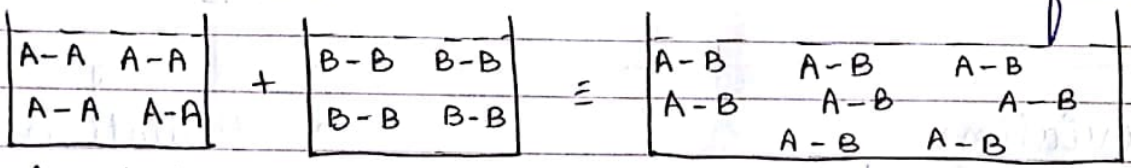
$P_B^\circ = 135$ mm of Hg

Ideal solution :-

Soln which follow Raoult's law at all Temp. & all concn, called ideal soln.

- ~~$\Delta H_{mix} \neq 0$~~ $\Delta H_{mix} = 0$ i.e. there is no change in heat during formn of soln

$\Delta V_{mix} = 0$ Vol. of soln = ~~Volw~~ Volm of solute + Volm of solvent



solvent (A) solute (B)

Forces of interaction b/w solute + Forces of interaction b/w solvent = Forces of interaction b/w solute & solvent particle

$A - A + B - B = A - B$

Note: No soln is ideal / perfect soln however sol'n which have same type of solute & solvent particles are ideal soln

Solute & solvent — n-butane + n-pentane ✓ (same nature)

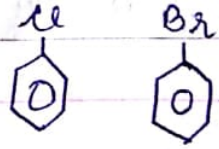


Benzene + Toulene

$CH_3-CH_2-Cl + CH_3-CH_2-Br_2$

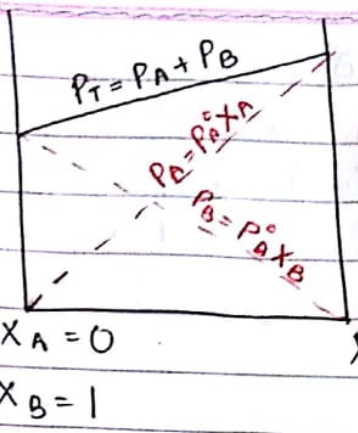
$CH_3-OH + C_2H_5OH$

$CCl_4 + SiCl_4$



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A → solvent
B → solute



$\Delta S = 0$

Note: Dilute sol^n are considered as ideal sol^n becoz amt. of solute is very less therefore effect of interaction b/w molecules (A-B) are ~~no~~ ~~negligible~~ neg

V. imp

Real sol^n / Non-Ideal sol^n :-

→ Sol^n which does not follow Raoult's law.

Observed V.P. (exp.) ≠ calculated V.P. (Theo.)

Observed B.P. ≠ Calculated B.P.

+ve Deviation

- observed V.P. > Calculated V.P.
- $\Delta H_{mix} > 0$
- $\Delta V_{mix} > 0$
- A-A & B-B > A-B
- Obs. BP < Calc. B.P.

eg CCl_4 & $CHCl_3$

CCl_4 & C_2H_5OH

Benzene & CCl_4

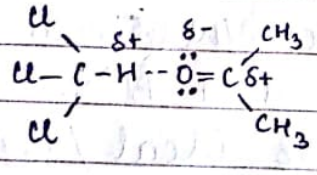
*C_2H_5OH & $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

C_2H_5OH & H_2O

Phenol & CS_2

Benzene & acetone

$\Delta S > 0$



−ve Deviation

Observed VP < Calc. V.P.

$\Delta H_{mix} < 0$

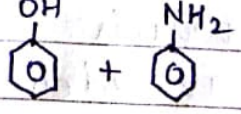
$\Delta V_{mix} < 0$

A-A & B-B < A-B

Obs. B.P. > Cal. B.P.

$CHCl_3 + CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

H_2O & HNO_3



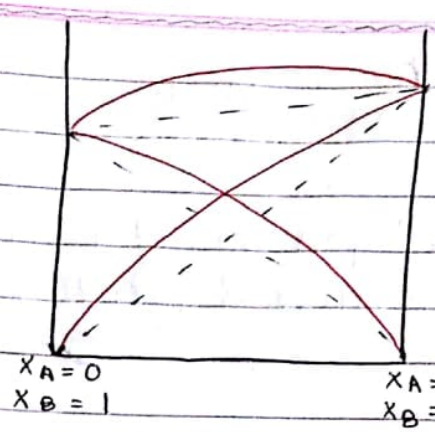
$H_2O + HCl$

$CHCl_3 + CH_3COOH$

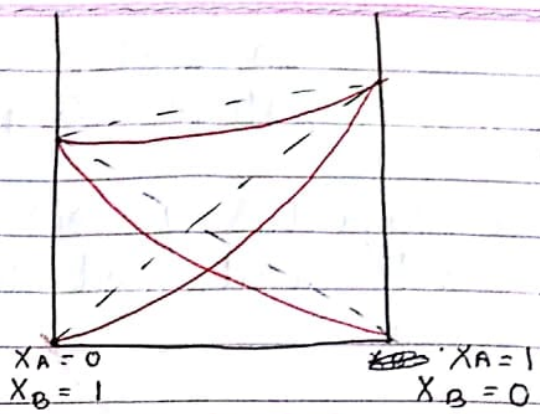
$CHCl_3 +$ 

$\Delta S < 0$

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+ve deviation



−ve deviation

Azeotrophic Mixture :- These are constant Boiling Mixture.

Azeotrophs are formed by mixing 2 different liquid in a fixed proportion so that the whole mixture boils at const. boiling temp.

★ Azeotrophs can't be separated by fractional distillation.

→ Fractional Distillation :-
Method of separating 2 liquids from their mixture on the basis of their different B.P.

Two types of Azeotrophics :-

Min^m Boiling azeotrophs	Max^m Boiling azeotrophs.
eg : +ve deviation real sol^n	-ve deviation real sol^n.
eg : C_2H_5OH + H_2O	eg H_2O + HNO_3
95% 5%	32% 68%

B.P. of azeotrophs < B.P. of mix. component

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Colligative Properties :-

solvent → volatile → A] V.P. ↓
solute → non volatile → B

The properties of soln which depend upon no. of particles in soln and not depend upon nature & identity of solute.

1. Relative lowering in V.P.
2. ~~Elevation~~ Elevation in B.P.
3. Depression in Freezing Point
4. Osmotic Pressure

These all 4 properties are connected with V.P.

1. Relative Lowering in V.P. :-

solvent → volatile → A → N moles] V.P. ↓
solute → non volatile → B → n-moles
↳ $P^\circ_B = 0$

soln

Acc^n to Raoult's law

$$P_T = P_A + P_B$$

$$P_T = P^\circ_A X_A + P^\circ_B X_B \quad (P^\circ_B X_B \to 0)$$

$$P_T = P^\circ_A X_A$$

$$P_T = P^\circ_A (1 - X_B)$$

$$\frac{P^\circ_A - P_T}{P^\circ_A} = X_B$$

$$\frac{P^\circ_A - P_T}{P^\circ_A} = \frac{n}{n+N}$$

relative lowering of V.P. ←

$$\boxed{\frac{\Delta P}{P^\circ_A} = \frac{n}{n+N}}$$

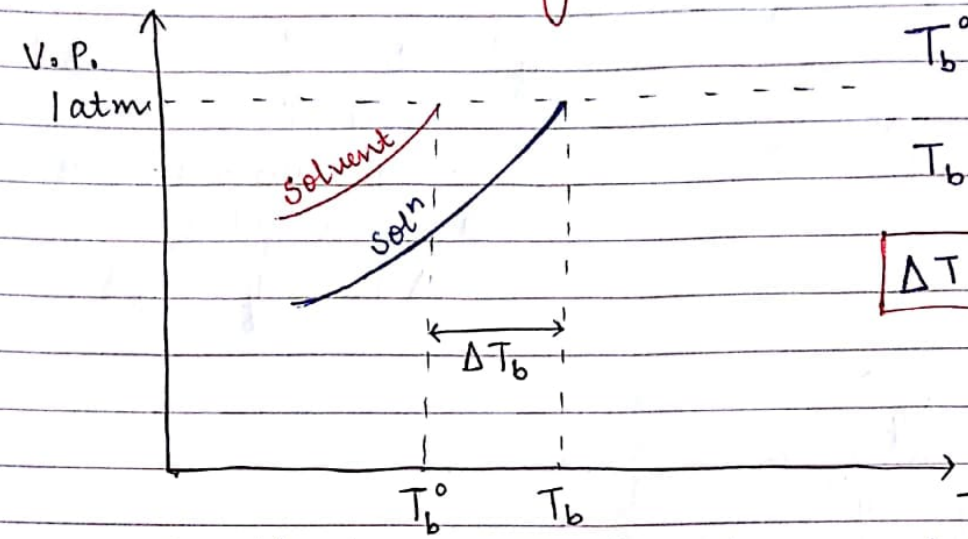
dilute soln $\quad n + N \approx N$

$$\boxed{\frac{\Delta P}{P^\circ_A} = \frac{n}{N}}$$

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$$\frac{\Delta P}{P^\circ_A} = \frac{W_B / M_{W_B}}{W_A / M_{W_A}}$$

2. Elevation in Boiling Point (ΔT_b) :-



T_b° → B.P. of pure solvent

T_b → B.P. of soln

$$\Delta T = T_b - T_b^\circ$$

ΔT_b is directly proportional to molality

$\Delta T_b \propto m$

$$\Delta T_b = K_b m$$

K_b → Elevation const. / Molal Elevation const or Ebullioscopic const.

Unit of K_b → K kg mol^{-1}

$$m = \frac{n_{solute}}{W_{solvent}\ (in\ kg)}$$

$$\Delta T_b = K_b \times \frac{W_B / M_B}{W_A / 1000}$$

$$\Delta T_b = K_b \times \frac{W_B \times 1000}{M_B \times W_A}$$

To calculate molar mass of solute ←

$$M_B = \frac{K_b \times W_B \times 1000}{\Delta T_b \times W_A}$$

$$K_b = \frac{R \times M_A \times T_b^{\circ 2}}{1000 \times \Delta_{vap}H}$$

R → Universal gas const.

M_A → molar mass of solvent

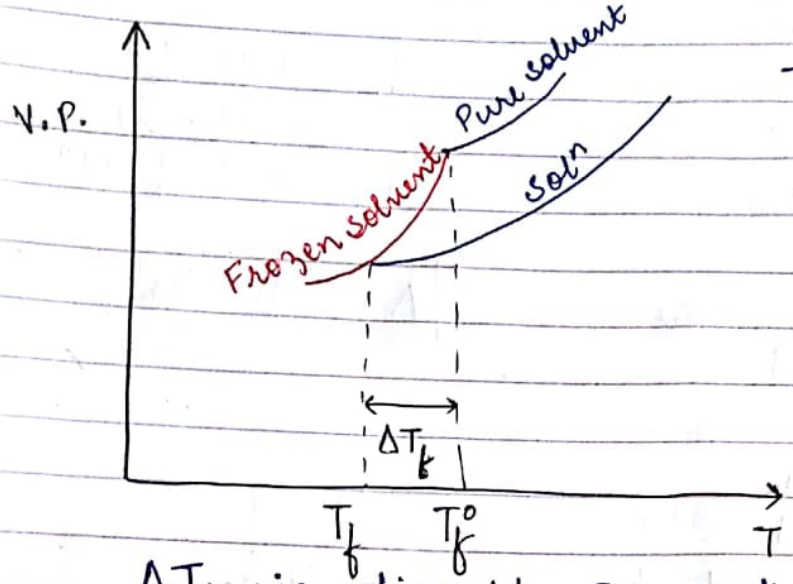
T_b° → B.P. of pure solvent

$\Delta_{vap}H$ → Heat of vap.

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3. Depression in Freezing Point (ΔT_f)

Freezing Point : The temp. at which vapour Pressure of solvent in liq. phase is same that of V.P. in solid ~~sol~~ state.



T_f° → F.P. of Pure solvent in liq. Phase

T_f → F.P. when non-volatile solute is added in it

$$\Delta T_f = T_f^\circ - T_f$$

ΔT_f is directly proportional to molality

$$\Delta T_f \propto m$$

$$\Delta T_f = K_f m$$

K_f → Depression const / molal Depression const (~~Cryo~~ Cryoscopic const.)

Unit of K_f :→ K kg mol^{-1}

$$\Delta T_f = K_f \times \frac{W_B / M_B}{W_A / 1000}$$

$$\Delta T_f = K_f \times \frac{W_B \times 1000}{M_B \times W_A}$$

To calculate molar mass of solute ←

$$M_B = \frac{K_f \times W_B \times 1000}{\Delta T_f \times W_A}$$

$$K_f = \frac{R \times M_A \times T_f^{\circ 2}}{1000 \times \Delta_{fusion} H}$$

R → Universal gas const.

M_A → molar mass of solvent

T_f° → F.P. of pure solvent

ΔH_{fusion} → heat of fusion

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$C_6H_{12}O_6$

$72 + 12 + 6 \times 16$

$84 + 96 = 180$

NCERT

Ques 36 g of glucose is dissolved in 500 gm water. At what temp. will be water boil? (K_b) water = 0.52 K Kg mol^{-1}

$W_B = 36\,g$
$W_A = 500\,g$
$K_B = 0.52$

$$\Delta T_b = \frac{K_b \times W_B \times 1000}{M_B \times W_A}$$

$$= \frac{0.52 \times 36 \times 1000}{180 \times 500}$$

$$= \frac{104}{5 \times 100}$$

$$\Delta T_b = 20.8 \times 10^{-2}$$

$\Delta T_b = T_b - T_b^\circ$

$\Delta T_b + T_b^\circ = T_b$

$0.208 + 373 = T_b$

$T_b = 373.208$

Ques The B.P. of Benzene is 353.23 K when 1.80 g non-volatile solute was dissolved in 90 g of Benzene. The B.P. is raised to 354.11 K. Calc. molar mass of solute K_b for benzene = 2.53 K kg mol^{-1}

$$M_B = \frac{K_b \times W_B \times 1000}{\Delta T_b \times W_A}$$

$$= \frac{2.53 \times 180 \times 1000}{0.88 \times 90}$$

$$= \frac{2530}{44} = 57.\ gm$$

354.11
353.23
0.88

$44 \overline{)2530}$
220
88
308

Ques B.P. of water at 750 mm of Hg is 99.63°C ~~How~~ How much sucrose is to be added to 500 gm of water such that it boils at 100°C

$\Delta T_b = T_b - T_b^\circ$

$= 100 - 99.63 = 0.37°C$

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$$\Delta T_b = K_b m$$

$$\Delta T_b = K_b \times \frac{W_B/M_B \times 1000}{W_A}$$

$$0.37 = 0.52 \times \frac{W_B \times 1000}{342 \times 500} \times 2$$

$$\frac{0.37}{0.52} \times \frac{342}{2} = W_B$$

$$W_B = \frac{63.27}{0.52} = 121.67 \text{ kg}$$

$72 + 8 + 96 = 80 + 96 = 176$

Ques Calc. Mass of Ascorbic Acid (Vitamin C, $C_6H_8O_6$) to be dissolved in 75 gm of acetic acid to lower its M.P. by 1.5°C ($K_f = 3.9$ K Kg mol^{-1})

$$M_B = \frac{K_f \times W_B \times 1000}{\Delta T_f \times W_A}$$

$$W_B = \frac{M_B \times \Delta T_f \times W_A}{K_f \times 1000}$$

$$= \frac{176 \times 1.5 \times 75}{3.9 \times 1000}$$

$$W_B = 5.076 \text{ gm}$$

Ques A sol^n of sucrose (Molar Mass = 342 gmol^{-1}) has been prepared by dissolving 68.5 gm of sucrose in 1000 gm of water. The F.P. of sol^n will be - (water → $K_f \approx 1.86$ K Kg mol^{-1})

① −0.372 °C ② −0.520 °C ③ +0.372°C ④ −0.570°C

$$\Delta T_f = \frac{K_f \times W_B \times 1000}{M_B \times W_A}$$

$$= \frac{1.86 \times 68.5 \times 1000}{342 \times 1000}$$

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$\Delta T_f = 0.3725$

$\Delta T_f = T_f^\circ - T_f$

$T_f = T_f^\circ - \Delta T_f$

$= 273 - 0.3725$

$T_f = 272.6275\ K$

AIPMT

Ques A soln of urea (Molar Mass = 56 gm mol^{-1}) boils at 100.18°C at atm. pressure. If K_f & K_b for water are 1.86 K kg mol^{-1} & 0.512 K kg mol^{-1} resp. the above soln will freeze at.

① −6.54°C ② 6.54°C ③ 0.654°C ④ −0.654°C

$$\Delta T_f = \frac{K_f \times W_B \times 1000}{M_B \times W_A}$$

$$\Delta T_b = \frac{K_b \times W_B \times 1000}{M_B \times W_A}$$

$$\frac{\Delta T_f}{\Delta T_b} = \frac{K_f}{K_b}$$

$$\frac{\Delta T_f}{100.18-100} = \frac{1.86}{0.512}$$

$$\Delta T_f = \frac{1.86}{0.512} \times 0.18 = \frac{0.3348}{0.512}$$

$= 0.6539°C$

$\Delta T_f = T_f^\circ - T_f$

$T_f = T_f^\circ - \Delta T_f$

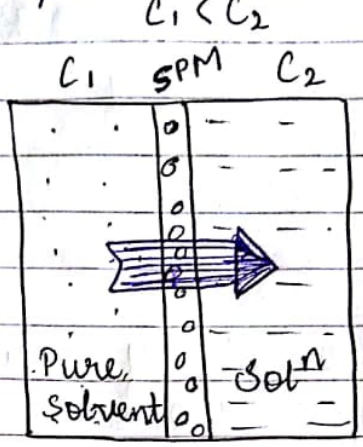
$= 273 - 0.6539$

$= 272.347\ K$

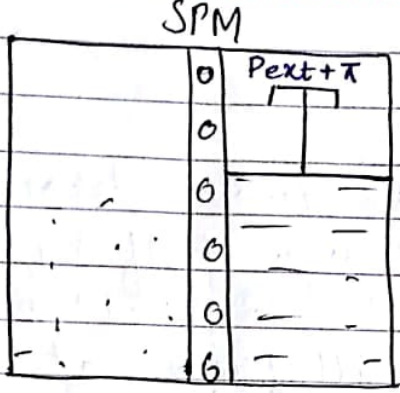
$= -0.653°C$

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4. Oso ⊗ Osmosis :- The flow of liquid from ~~low~~ dilute sol^n (low conc^n) to concentrated sol^n (high conc^n) through semi permeable membrane is c/a osmosis.



Osmotic Pressure (π):

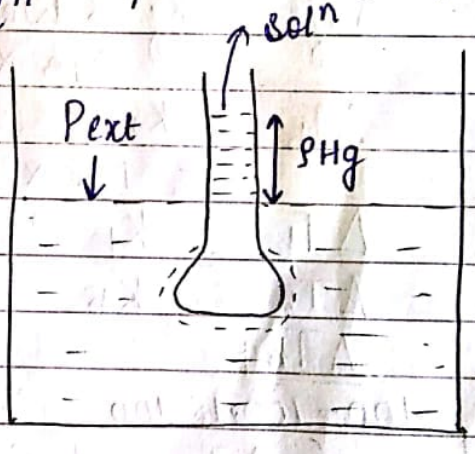


The min^m ext. pr. required to apply on sol^n side to prevent osmosis, is c/a osmotic pressure (π)

Osmotic pr. $\propto$ conc. of sol^n

$$\pi \propto C$$
$$\pi = CRT$$
$$\pi = n_B/V \, RT$$
$$\pi = \frac{W_B/M_B \, RT}{V}$$

Formula to calculate Mw to solute. ← $$M_B = \frac{W_B RT}{\pi V}$$



AIPMT This method is used to find M.W. of biomolecules like proteins, polymers & other macromolecules becoz they are not stable at high temp.

Type of Solution :-

1. Isotonic sol^n : Two sol^n have same osmotic pressure at a given temp. c/a isotonic sol^n.

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When these 2 sol^n are separated by SPM there is no osmosis.

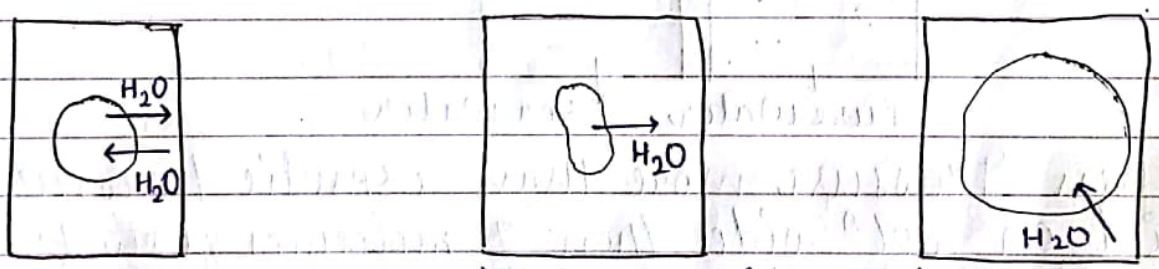
2. Hypertonic & Hypotonic sol^n :-
Osmotic pr. of blood $\approx$ 0.9 % $(\frac{w}{v})$ of NaCl

Hypertonic sol^n :-
sol^n which have conc^n more than 0.9 % w/v NaCl

→ When cell is placed in such sol^n water will flow out of cell and cell would ~~sink~~. shrink.

Hypotonic sol^n :-
sol^n which have conc^n less than 0.9 % w/v NaCl.

→ When cell is placed in such sol^n water will flow inside the cell and cell would swell up.



conc. $\approx$ 0.9 % w/v NaCl → isotonic

conc. > 0.9 % w/v NaCl → hypertonic, cell shrink

conc^n < 0.9 % w/v NaCl → Hypotonic sol^n

Edema :- When a person take a lot of salt or salty food water comes out of RBC & water ~~retention~~ retention in tissue take place which cause intracellular space resulting in swelling, ~~this~~ this is known as Edema.

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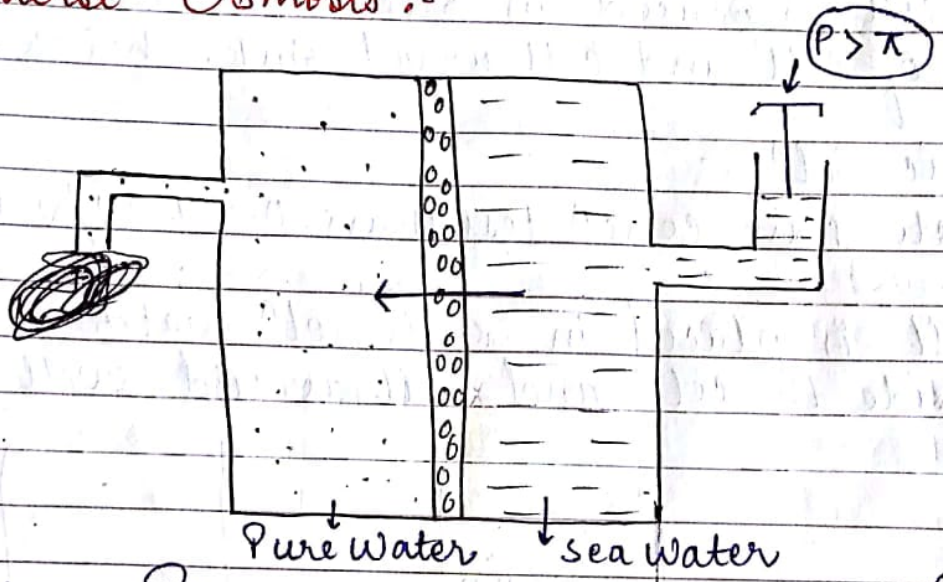
Other Example of Osmosis :-

- Raw mangoes placed in salty water loses water via osmosis & shrival into pickles.
- Wilted flower revive when placed in water

Isohydric Solutions :-

2 sol^{n}s which have same H^+ conc^n i.e. same pH c/a isohydric sol^n.

Reverse Osmosis :-



When Pressure more than osmotic Pressure is applied on sol^n side then solvent molecules flow from sol^n to solvent. This process is k/n as reverse osmosis.

AIPMT

Que: A 5% sol^n of cane sugar (Mw = 342) is isotonic with 1% sol^n of a substance X. The Mol. wt. of X is –

(1) 34.2 (2) 171.2 (3) 68.4 (4) 136.8

$\pi_1 = \pi_2$

$$\frac{5/342}{100} RT = \frac{1/M}{100} RT$$

$$\Rightarrow \frac{5}{342} = \frac{1}{M}$$

$$M = \frac{342}{5} = 68.4$$

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M = 150

Que A soln is prepared by 10 g of an org. compd in one litre soln at 27°C osmotic pr. of soln is 1.64 atm. Calculate molar mass of solvent

$$\pi = CRT$$

$$1.64 = \frac{W/M}{V} RT$$

$$M = \frac{W RT}{V \times 1.64}$$

$$= \frac{5}{1} \times \frac{0.082 \times 300 \times 100}{1 \times 164}$$

$$M = 150$$

Vant Hoff factor (i) :-

Colligative Properties depends upon no. of particle.

$KCl \rightleftharpoons K^+ + Cl^-$ — dissociation

$2CH_3COOH \rightleftharpoons (CH_3COOH)_2$ — Association

60 g → 120 g

$$i = \frac{\text{Observed C.P.}}{\text{Calculated C.P.}}$$

$$i = \frac{\text{normal molar mass}}{\text{abnormal mass}}$$

$$i = \frac{\text{no. of particles after dissociation / association}}{\text{no. of particles before dissociation / association}}$$

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Dissociation :-

no. of Particles ↑

$i > 1$ ⇒ Obs. C.P. > Calc. C.P.

normal molar mass > abnormal molar mass

$$A_xB_y \rightleftharpoons x A^{y+} + y B^{x-}$$

1	0	0
$1-\alpha$	$x\alpha$	$y\alpha$

Total mole after dissociation $= 1-\alpha+x\alpha+y\alpha$

$= 1-\alpha+(x+y)\alpha$

$= 1-\alpha+n\alpha$

$x+y=n$

n → Total dissociation ion moles

$$i = \frac{\text{total moles after dissociation}}{\text{total moles before dissociation}}$$

$$i = \frac{1-\alpha+n\alpha}{1}$$

$$i = 1+(n-1)\alpha$$

Calculation of DOD ← $\boxed{\alpha = \frac{i-1}{n-1}}$

For strong electrolyte

DOD = 100 % ⇒ $\alpha = 1$

$$1 = \frac{i-1}{n-1}$$

$\boxed{i = n}$

strong electrolyte

- $NaCl \rightarrow Na^+ + Cl^-$ $\quad i = 2$
- $K_2SO_4 \rightarrow 2K^+ + SO_4^{2-}$ $\quad i = 3$

$$K_4[Fe(CN)_6] \rightleftharpoons 4K^+ + [Fe(CN)_6]$$

(1) If $\alpha = 100\%$ $\quad \alpha = \frac{i-1}{n-1}$

$i = n$

$i = 5$

(2) If $\alpha = 50\%$ $\quad \frac{50}{100} = \frac{i-1}{5-1}$

$5-1 = 2i-2$

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$4 = 2i - 2$

$i = 3$

Observed C.P. = i Calc. C.P.

$\frac{\Delta P}{P^\circ} = i \times \frac{n}{N}$ $\qquad \pi = i\,CRT$

$\Delta T_b = i \times K_b m$ $\qquad \pi = i \frac{n_B}{V} RT$

$\Delta T_f = i \times K_f m$

Association : $i < 1$

obs. C.P. < Calc. C.P.

normal molar mass < abnormal molar mass

nA	$\rightleftharpoons$	A_n
1		0
$1-\alpha$		α/n

Total moles after association $= 1 - \alpha + \alpha/n = 1 - \left(1 - \frac{1}{n}\right)\alpha$

Total moles before association = 1

$$i = \frac{1 - (1 - \frac{1}{n})\alpha}{1}$$

$$\alpha = \frac{1 - i}{1 - \frac{1}{n}}$$

DOD = 100%.

$$\alpha = 1 = \frac{1 - i}{1 - \frac{1}{n}}$$

$$1 - i = 1 - \frac{1}{n}$$

$$i = \frac{1}{n}$$

$2\,CH_3COOH \rightleftharpoons (CH_3COOH)_2$

$i = \frac{1}{2}$

AIPMT-11

Ques The F.P. depression const. for water is $-1.86\,^\circ C\,m^{-1}$. If 5.00 g Na_2SO_4 is dissolved in 45.0 g H_2O. The F.P. is changed by $-3.82\,^\circ C$. Calculate the van't Hoff factor for Na_2SO_4.

$\Delta T_f = i K_f m$

273 / −3.82 / 1.91

273 / −3.82 / 269.18

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$Na_2SO_4 \rightarrow 46 + 32 + 64$
$= 142$

$+3.82 = i \times (+1.86) \times \frac{5}{142} \times 1000$ / 45

$3.82 \times 45 \times 142 = i$
$1.86 \times 5 \times 1000$

$i = \frac{382 \times 3 \times 71}{31 \times 1000} = \frac{381 \times 3 \times 71}{31 \times 1000}$
≈ 2.7051

Ques 0.5 molal aq. soln of a weak acid HX is 20% ionised. If K_f for water is 1.86 Kg mol^{-1} The lowering in F.P. of soln is -

① -1.12 K ② 0.56 K ③ 1.12 K ④ -0.56 K

$HX \rightarrow H^+ + X^-$

$\alpha = \frac{i-1}{n-1} = 0.20 = \frac{i-1}{2-1}$

$0.20 = \frac{i-1}{1}$

$0.20 + 1 = i$

$i = 1.20$

$\Delta T_f = i \times K_f m$
$= 1.20 \times 1.86 \times 0.5$
$= 0.600 \times 1.86$
$\Delta T_f = 1.116\ ^\circ C$

Ques A 0.0020 m aq. soln of an ionic compd. $Co(NH_3)_5(NO_2)Cl$ Freeze at $-0.00732\ ^\circ C$. No. of moles of ion which 1 mol of ionic compd. produces on being dissolved in water will be ($K_f = -1.86\ ^\circ C/m$)

① 2 ② 3 ③ 4 ④ 1

ΔT

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Que • 2.925 g of NaCl is dissolved in 100 g water. Determine BP of sol^n (K_b for water = 0.52 K Kg mol^{-1})

$\Delta T_b = K_b m$

$T_b - 373 = 0.52 \times \frac{\frac{2.925}{58.5}}{100} \times 1000$

$T_b = \frac{0.52 \times 2.925 \times 1000}{100 \times 58.5 \times 100}$

$= \frac{52 \times 5 \times 10}{1000} = 0.2$

Calc. % DOD of an electrolyte XY_2 (Normal Mw = 164) in water. If observed Mw by measuring elevation in bp is 0.56.

$i = 0.56$

$\alpha = \frac{i-1}{n-1}$ $\qquad n = 1+2 = 3$

$\alpha\% = \frac{0.56-1}{3-1} \times 100\%$

=

Que F.P. of 0.01M aq. sol^n of $K_3[Fe(CN)_6]$ is -0.062 °C what is α for this complex. If K_f for water = 1.26 Kg / mol^{-1})

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$\Delta T_f = i K_f m$

$0 - (-0.062) = i \times 1.26 \times 0.01$

$\frac{0.062}{1.26 \times 0.01} = i$

$i = 4.9 \approx 5$

$\alpha = \frac{i-1}{n-1}$

$\alpha = \frac{5-1}{4-1} = \frac{4}{3}$

Ques 1 molal carboxyllic acid in benzene shows a elevation in B.P. of 1.518 K. Calculate Degree of dissociation for dimerisation of acid in benzene K_b for benzene = 2.53 K Kg mol^{-1}

$\Delta T_b = i K_b m \Rightarrow i = \frac{\Delta T_b}{K_b m} = \frac{1.518}{2.53 \times 1} = 0.6$

$2\,CH_3COOH \rightleftharpoons (CH_3COOH)_2$

1	0
$1-\alpha$	$\alpha/2$

$i = \frac{1 - \alpha/2}{1}$

$0.6 = 1 - \frac{\alpha}{2}$

$\frac{\alpha}{2} = 1 - 0.6 = 0.4$

$\alpha = 0.8$

Ques Calculate the osmotic pr. of 20% W/V anhy. $Ca(NO_3)_2$ soln at 0°C assuming 80% dissociation.

$Ca(NO_3)_2 \rightleftharpoons Ca^{2+} + 2NO_3^-$

1		
$1-\alpha$	α	2α

$i = \frac{1 - \alpha + \alpha + 2\alpha}{1} = \frac{1 + 2\alpha}{1} = 1 + 2 \times 0.8$

$= 1 + 1.6 = 2.6$

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$Ca(NO_3)_2 = 40 + 28 + 6\times16$
$= 68 + 96$
$= 164$

$\pi = iCRT$
$= \frac{W/M}{V} RT \times i$

20 gm solute + water in 100 ml soln

$= \frac{20}{164} \times \frac{1000}{100} \times 0.082 \times 273 \times 2.6$

$\pi = 27.30 \times 2.6$

$=$

Ques 2 soln of glucose have osmotic pr 1.5 atm & 2.5 atm resp. 1 L of 1st soln is mixed with 2 L of IInd soln. Find osmotic Pr. of resultant soln.

$\pi = iCRT$
$= i \frac{n_{BT}}{V_T} RT$

$\pi = i\frac{n}{V} RT$

$\frac{\pi V}{RT} = in$

$\pi V_T = (in)_{Total} RT$
$= \left(\frac{\pi_1 V_1}{RT} + \frac{\pi_2 V_2}{RT}\right) RT$

$\pi V_T = \pi_1 V_1 + \pi_2 V_2$
$= 1.5 \times 1 + 2.5 \times 2$
$\pi \times 3 = 1.5 + 5.0 = 6.5$
$\pi = \frac{6.5}{3} = 2.1$

Ques Following soln are given arrange them in ↑ order of V.P., B.P., F.P. & O.P. soln.

① 1 M sugar ($i=1$) ② 1 M Na_2SO_4 ($i=3$) ③ 1 M KCl ($i=2$) ④ 1 M $Al_2(SO_4)_3$ ($i=5$)

V.P. → 4 < 2 < 3 < 1
B.P. → 1 < 3 < 2 < 4
F.P. → 4 < 2 < 3 < 1
O.P. → 1 < 3 < 2 < 4

$\Delta T_b = T_b - T_b^\circ$
$\Delta T_b + T_b^\circ = T_b$
B.P. ↑

$\Delta T_f = T_f^\circ - T_f$
$T_f = T_f^\circ - \Delta T_f$
F.P. ↓

$P^\circ - P_T = \Delta P$
$P^\circ - \Delta P = P_T$
V.P. ↓

$\pi = iCRT$

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Ques Find max^n & min^m C.P. of following aq. sol^n.

① 1% W/V sugar ② 1% W/V urea ③ 1% W/V Na_2SO_4 ④ 1% W/V NaCl

$C = \frac{1/342 \times 1000}{100}$ $C = \frac{1/60 \times 1000}{100}$ $C = \frac{1/142 \times 1000}{100}$ $\frac{1/58.5 \times 1000}{100}$

$C = \frac{10}{342}$ $= \frac{10}{60}$ $= \frac{10}{142}$

$i \times C = 0.029$ $= \frac{1}{6} = 0.16$ $= 3 \times \frac{10}{142} =$ $= 0.34$

↓ min^m $= 0.211$ ↓ max

C.P. $\propto$ Observed conc^n ($i \times c$)

AIPMT-14

Ques which of the following sol^n will exhibit the largest F.P. depression?

① KCl ② $C_6H_{12}O_6$ ③ $Al_2(SO_4)_3$ ④ K_2SO_4

$i = 2$ $i = 1$ $i = 5$ $K = 3$

$$\Delta T_f = i K_f m$$

Ques An aq. sol^n is 1.00 m in KI which change will cause rise in V.P. of sol^n.

① add^n of NaCl ② Add^n of Na_2SO_4 ③ Add^n 1.00 molal KI

④ add^n of water

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Ques Which of the following aq. soln have min^m F.P.?

(1)	0.01 M NaCl	$i = 2$	$T_f = T_f^\circ - 0.02\ K_f$
(2)	0.005 m MgI_2	$i = 3$	$T_f = T_f^\circ - 0.015\ K_f$
(3)	0.005 m C_2H_5OH	$i = 2$	$T_f = T_f^\circ - 0.010\ K_f$
(4)	0.005 m $MgSO_4$	$i = 2$	$= T_f^\circ - 0.010\ K_f$

$$\Delta T_f = T_f^\circ - T_f$$

$$T_f = T_f^\circ - \Delta T_f$$

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